MATH 006 – DISCRETE MATHEMATICS

FINAL – LESSON 2: COMBINATIONS

Recall: Counting Techniques

A farmer is selecting one goat and one sheep. If there are 5 goats and 8 sheep available, in how many ways can the farmer make his purchase?

By MP: $5 \times 8 = \underline{40}$

Let’s make it complicated…

Suppose another farmer wants to buy 2 pigs and 3 sheep.

How are we going to count the number of ways that this farmer can make a purchase?

Combination

- Let k, n be nonnegative integers such that $k \leq n$.
- The number of ways to select k objects from a set of distinct n objects is:

$$C(n,k) \text{ or } \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

$n! = n(n-1)(n-2)\cdots(2)(1)$

$0! = 1$

- The objects enumerated by the above number is called k-combinations.

Some Basic Properties

$$\binom{n}{0} = 1$$

$$\binom{n}{1} = n$$

$$\binom{n}{n} = 1$$

$$\binom{n}{k} = \binom{n}{n-k}$$

Example 1

A farmer buys 3 cows, 2 pigs, and 4 hens from aman who has 6 cows, 5 pigs, and 8 hens. Find the number of choices that the farmer has.

By MP: $\binom{6}{3} \times \binom{5}{2} \times \binom{8}{4} = \underline{14000}$ //

Example 2

A class is consists of 10 students with 6 men and 4 women. Find the number of ways to form a 4-member committee with:

1. no restrictions $\binom{10}{4} = 210$

2. equal number of men and women

   By MP: $\binom{6}{2} \times \binom{4}{2} = 90$

3. more women than men

Example 2

3.) Case 1: 3W, 1M

$\binom{4}{3} \times \binom{6}{1} = 24$

Case 2: 4W

$\binom{4}{4} = 1$

By AP:

$\underline{25}$

Example 3

A box contains 8 blue balls, 6 red balls, and 4 blackballs. Four balls are to be drawn successively without replacement, compute the number of possibilities of the following combinations:

1. all possible combinations $\binom{18}{4}$
2. all blue balls $\binom{8}{4}$
3. equal number of blue and black balls $\binom{8}{2} \times \binom{4}{2}$
4. no red balls $\binom{12}{4}$
5. at least 2 black balls

≥ 2

Example 3

3.) Case 1: 2 Blacks, 2 any color

$$\binom{4}{2} \times \binom{14}{2} = 546$$

case 2: 3 Blacks, 1 any color

$$\binom{4}{3} \times \binom{14}{1} = 56$$

case 3: 4 Blacks

$$\binom{4}{4} = 1$$

By AP:

$\underline{603}$ //

Example 4

In flipping 5 coins, how many possible outcomes are there

1. in total? $\binom{2}{1}\binom{2}{1}\binom{2}{1}\binom{2}{1}\binom{2}{1} = 2^5 = 32$
2. that has 2 heads? HHTTT, THTTH $\binom{5}{2} = 10$
3. with at most 3 tails?

≤ 3

Case 1: 3T $\binom{5}{3} = 10$

Case 2: 2T $\binom{5}{2} = 10$

Case 3: 1T $\binom{5}{1} = 5$

Case 4: 0T $\binom{5}{0} = 1$

By AP: 26

Example 5

How many bit strings with length 6 are there with at least 3 zeros?

D. I. Y.